

Essential Summary

A Modified Perturbation Method for Global Dynamic Analysis of Generalized Mixed Rayleigh–Liénard Oscillator with Cubic and Quintic Nonlinearities

Zhenbo Li, Linxia Hou, Yiqing Zhang, Feng Xu

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1 Background and Motivation

Deriving an analytical relationship between system parameters and the amplitude of limit cycles is a meaningful yet challenging task. When the restoring force or nonlinear damping is complicated, many existing analytical methods **cannot execute their procedures symbolically** — they require all system parameters to be assigned numerically upfront, yielding only isolated snapshots in parameter space rather than a continuous global picture.

This work proposes a **modified generalized harmonic function perturbation method** and applies it to a generalized mixed Rayleigh–Liénard oscillator with both cubic and quintic nonlinearities. This paper is historically significant as the **first publication** introducing the composite Simpson integration modification to the GHFP framework. The proposed method enables the analytical derivation of amplitude–parameter relationships, allowing quantitative analysis of the existence, stability, number, position, and amplitude of each limit cycle.

2 System Model

The generalized mixed Rayleigh–Liénard oscillator studied (Eq. 3 of the original paper):

$$\ddot{x} + c_1x + c_3x^3 + c_5x^5 = \varepsilon(\mu + \mu_1x + \mu_2x^2 + \mu_3x^3 + \mu_4x^4 + \mu_{22}\dot{x}^2)\dot{x} \quad (1)$$

where c_i are stiffness parameters (restoring force includes cubic and quintic terms), μ is the control parameter, and μ_i, μ_{22} are damping coefficients combining Liénard-type (x, x^2, x^3, x^4) and Rayleigh-type ($\dot{x}^2$) damping.

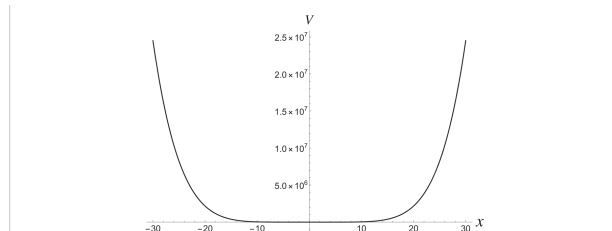


Figure 1: **Figure 1.** Potential portrait of the oscillator under single-well conditions.

3 The Modified GHFP Method

3.1 Key Innovation

The classical GHFP method fails for complicated oscillators because the integrals involving non-linear terms cannot be evaluated analytically. The **composite Simpson quadrature formula** is introduced to discretize these integrals:

$$\int_a^b f(x) dx \approx \frac{h}{3} \left[f(x_0) + 4 \sum_{i \text{ odd}} f(x_i) + 2 \sum_{i \text{ even}} f(x_i) + f(x_n) \right] \quad (2)$$

By setting $n = 18$ subintervals and $M = 4$ harmonic terms, the Fourier coefficients p_{2i} ($i = 0, 1, 2, 3, 4$) become **explicit algebraic functions** of the system parameters. The entire perturbation procedure can now be executed purely symbolically.

3.2 Procedure

1. Nonlinear time transformation $\frac{d\varphi}{dt} = \Phi(\varphi)$ with $\Phi(\varphi) = \Phi_0 + \varepsilon\Phi_1 + \dots$
2. Solution expansion $x = a \cos^2 \varphi + b$, with $a = a_0 + \varepsilon a_1 + \dots$
3. Perturbation expansion: zero-order for free oscillation, first-order for damping effects
4. Composite Simpson integration ($n = 18$) to compute Fourier coefficients p_{2i} analytically

4 Key Equations

4.1 Amplitude–Parameter Relation (Eq. 35)

The control parameter μ is expressed as a function of the limit cycle amplitude A and all system parameters:

$$\mu = \frac{4}{\pi a_0^2 (2p_0 - p_4)} \left(E_1 \mu_1 + E_2 \mu_2 + E_3 \mu_3 + E_4 \mu_4 + E_{22} \mu_{22} \right) \quad (3)$$

where p_{2i} are Fourier coefficients and the coefficient functions are:

$$\begin{aligned} E_1 &= -\frac{1}{4} \pi a_0^2 b (2p_0 - p_4) - \frac{1}{16} \pi a_0^3 (4p_0 + p_2 - 2p_4 - p_6) \\ E_2 &= -\frac{1}{64} \pi a_0^2 (8a_0 b_0 (4p_0 + p_2 - 2p_4 - p_6) + a_0^2 (10p_0 + 4p_2 - 4p_4 - 4p_6 - p_8) \\ &\quad + 16b_0^2 (2p_0 - p_4)) \\ E_3 &= -\frac{1}{256} \pi a_0^2 (12a_0^2 b_0 (10p_0 + 4p_2 - 4p_4 - 4p_6 - p_8) + 48a_0 b_0^2 (4p_0 + p_2 - 2p_4 \\ &\quad - p_6) + a_0^3 (28p_0 + 14p_2 - 8p_4 - 13p_6 - 6p_8) + 64b_0^3 (2p_0 - p_4)) \\ E_4 &= -\frac{1}{1024} \pi a_0^2 (16a_0^3 b_0 (28p_0 + 14p_2 - 8p_4 - 13p_6 - 6p_8) + 96a_0^2 b_0^2 (10p_0 + 4p_2 \\ &\quad - 4p_4 - 4p_6 - p_8) + 256a_0 b_0^3 (4p_0 + p_2 - 2p_4 - p_6) + a_0^4 (84p_0 + 48p_2 \\ &\quad - 15p_4 - 40p_6 - 26p_8) + 256b_0^4 (2p_0 - p_4)) \\ E_{22} &= -\frac{3}{64} \pi a_0^4 (8p_0^3 + 4(p_8 - 4p_4)p_0^2 + 2(2p_2^2 - 6p_6p_2 + 7p_4^2 - 8p_4p_8 + 6(p_6^2 + p_8^2))p_0 \\ &\quad - 4p_4^3 + p_8^3 - 7p_4p_6^2 - 8p_4p_8^2 + 8p_4^2p_8 - 2p_6^2p_8 - p_2^2(p_4 + 2p_8) \\ &\quad + 2p_2p_6(3p_4 + 2p_8)) \end{aligned}$$

For symmetric limit cycles: $a_0 = 2A$, $b = -A$. The relationship $\mu(A)$ is obtained purely symbolically — no numerical integration is required once the Fourier coefficients are computed.

4.2 Characteristic Quantity and Stability (Eq. 50)

$$h_0 = \frac{1}{\pi} \int_0^\pi \Phi_0 \left(\mu + \sum_{i=1}^4 \mu_i (a_0 \cos^2 \varphi + b)^i + 3\mu_{22} (-2a_0 \Phi_0 \cos \varphi \sin \varphi)^2 \right) d\varphi \quad (4)$$

where $h_0 = H_0/\varepsilon$. Based on the qualitative theory of ODEs:

- $h_0 > 0$: the limit cycle is **unstable**
- $h_0 = 0$: the limit cycle is **semi-stable** (bifurcation point)
- $h_0 < 0$: the limit cycle is **stable**

4.3 Homoclinic/Heteroclinic Bifurcation

If the initial amplitude A is set to the saddle point coordinate, Eq. (3) yields the critical value μ^{hom} or μ^{hetero} for homoclinic or heteroclinic bifurcation.

5 Example 1: Single-Well Potential (Section 4.1)

The oscillator (53) with parameters $c_1 = 1$, $c_3 = 1$, $c_5 = 0.2$, $\mu_1 = 0.1$, $\mu_2 = 1$, $\mu_3 = 0.1$, $\mu_4 = -1$, $\mu_{22} = 0.1$, $\varepsilon = 0.01$.

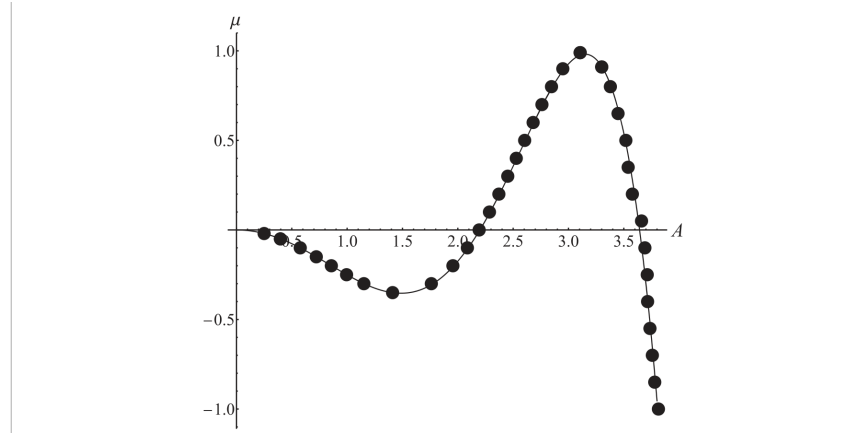


Figure 2: **Figure 4.** Comparison of μ - A curves ($\varepsilon = 0.01$). Solid: MGHFP. Dotted: Runge-Kutta.

Key observations:

- Below a critical μ : **no limit cycle exists**
- At the critical value ($\mu \approx -0.62$), a **semi-stable** limit cycle emerges
- As μ increases, it splits into stable ($h_0 < 0$) and unstable ($h_0 > 0$) cycles
- The unstable cycle eventually **collapses** to the equilibrium
- Up to **three coexisting limit cycles** at $\mu = -0.3$ (one unstable, two stable)

6 Example 2: Triple-Well Potential (Section 4.2)

The oscillator (66) with a richer potential landscape. The multiple wells enable both homoclinic and heteroclinic bifurcations.

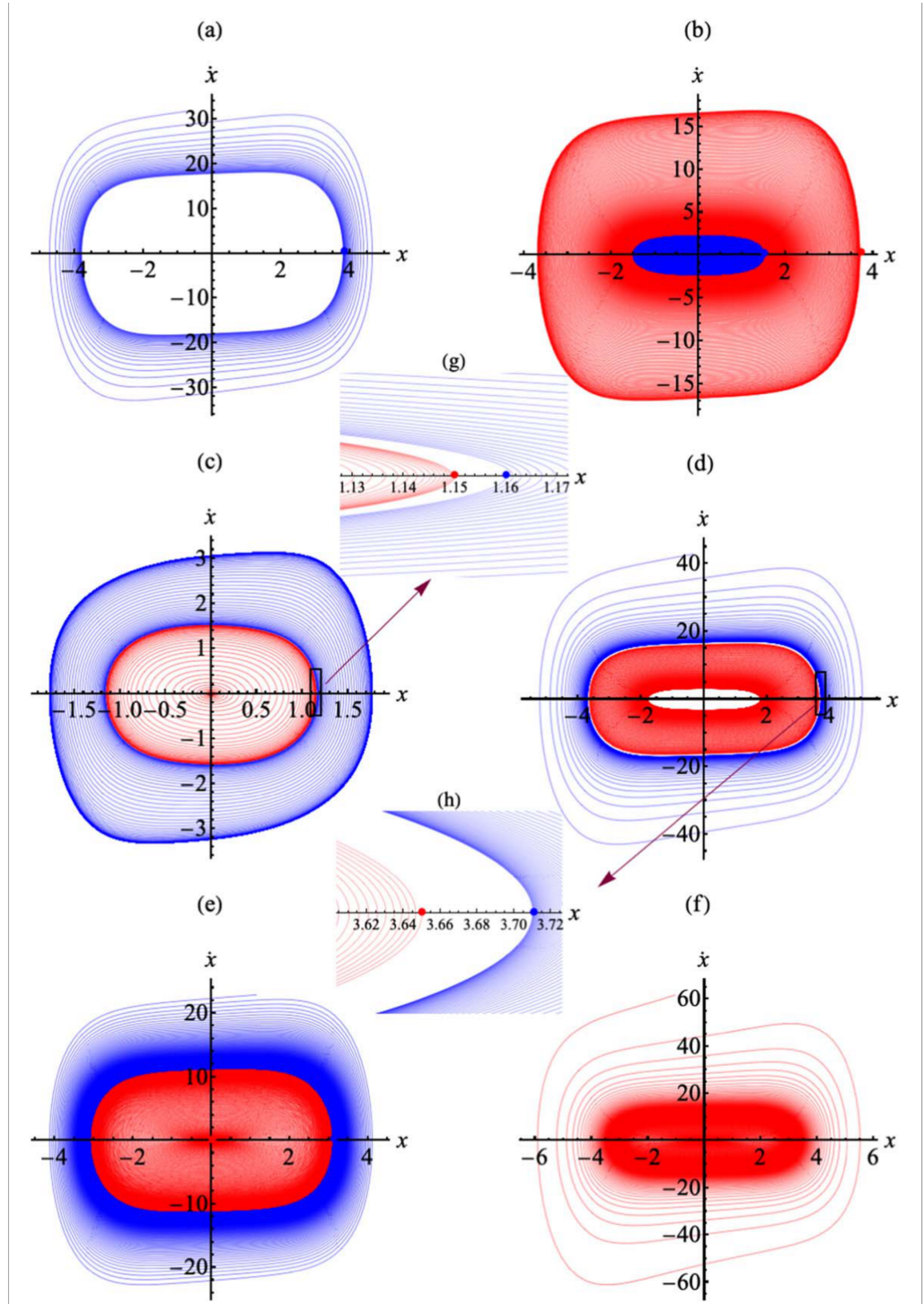


Figure 3: **Figure 3.** Phase portraits with different μ and A ($\varepsilon = 0.01$). Colorized dots: initial trajectory points.

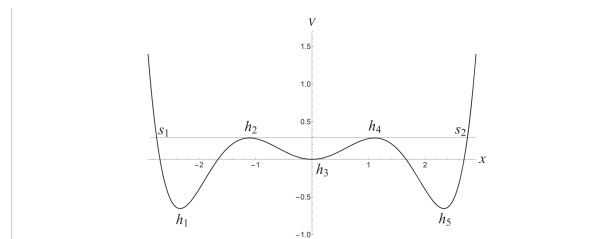


Figure 4: **Figure 6.** Triple-well potential portrait.

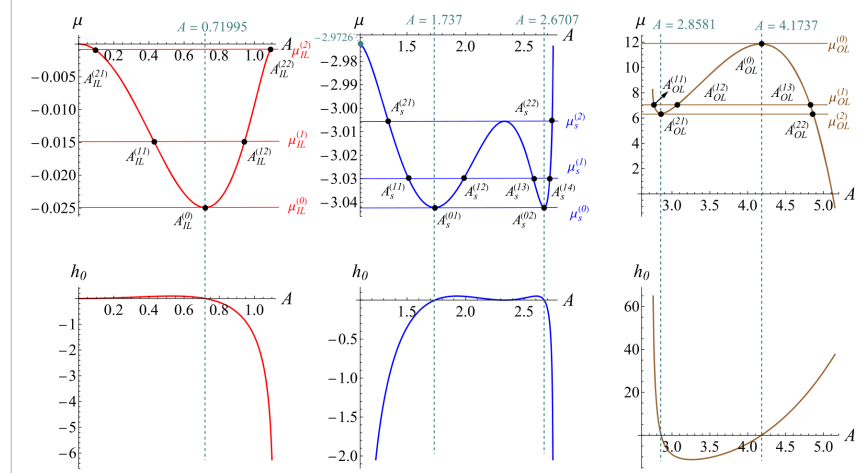


Figure 5: **Figure 7.** μ - A and h_0 - A curves for the triple-well oscillator.

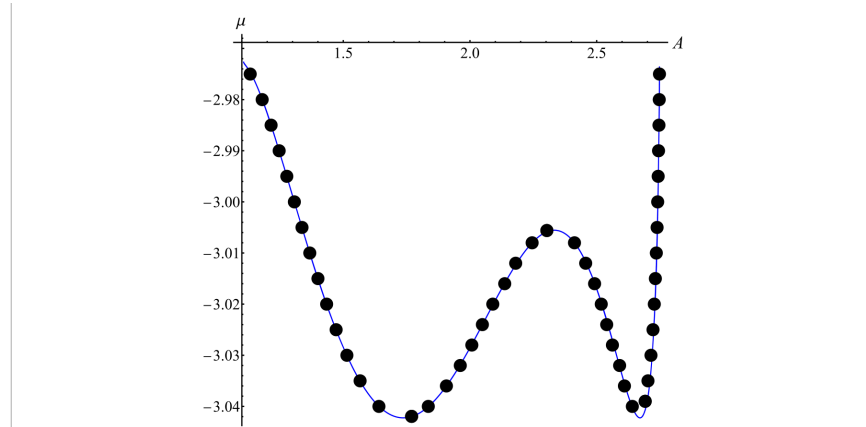


Figure 6: **Figure 9.** μ - A curves for limit cycles between saddle points ($\varepsilon = 0.1$). Method accurately predicts bifurcation thresholds.

7 Key Findings

1. **First MGHPF publication (Eqs. 35 & 50):** This is the first paper to introduce the composite Simpson formula into the GHFP framework. The analytical amplitude–parameter relation (Eq. 35) and stability criterion (Eq. 50) are derived purely symbolically without any parameter pre-assignment.
2. **Complete lifecycle prediction:** The μ – A relationship reveals the full lifecycle of each limit cycle — from generation through bifurcation to termination — covering stability, number, and amplitude.
3. **Up to three coexisting limit cycles:** Under certain conditions (e.g. $\mu = -0.3$), three limit cycles coexist simultaneously — a complex dynamical scenario that purely numerical methods struggle to fully capture.
4. **Homoclinic/heteroclinic bifurcation thresholds:** Both types of global bifurcation are quantitatively predicted. The analytical expressions serve as precursors to chaos.
5. **Dual engineering applications:** The framework supports both (i) limit cycle **amplitude regulation** (inertial impact shaker design, population dynamics) and (ii) **chaos prediction** via bifurcation thresholds (encryption, equipment protection).

8 Significance

This paper represents the **first publication** of the MGHPF methodology and compensates for the insufficiency of existing perturbation-based approaches. The method can be adopted for many other strongly nonlinear oscillators, including Duffing-harmonic oscillators and Helmholtz–Duffing oscillators.

Author: Zhenbo Li (lizhenbo@usc.edu.cn)
 University of South China, School of Mathematics and Physics
 Hunan Key Laboratory of Mathematical Modeling and Scientific Computing